

VAYU

Low Energy Passive Cooling System

Ventilation Assisted Yielding Utility

PK09: Passive Cooling of Building

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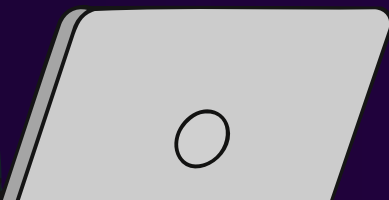
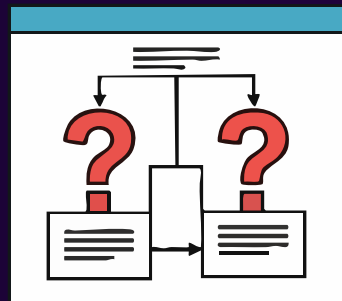
Sai Pragyan Rath

Siddha Saketh Ram Yarrabothu



The Problem

- Rising ambient temperatures are increasing the dependence on energy-intensive cooling systems.
- Conventional air conditioners consume significant electrical power and increase operational costs.
- Sustainable and affordable cooling solutions remain limited for many communities.
- Passive cooling strategies are often underutilized in modern building design.





Our Solution

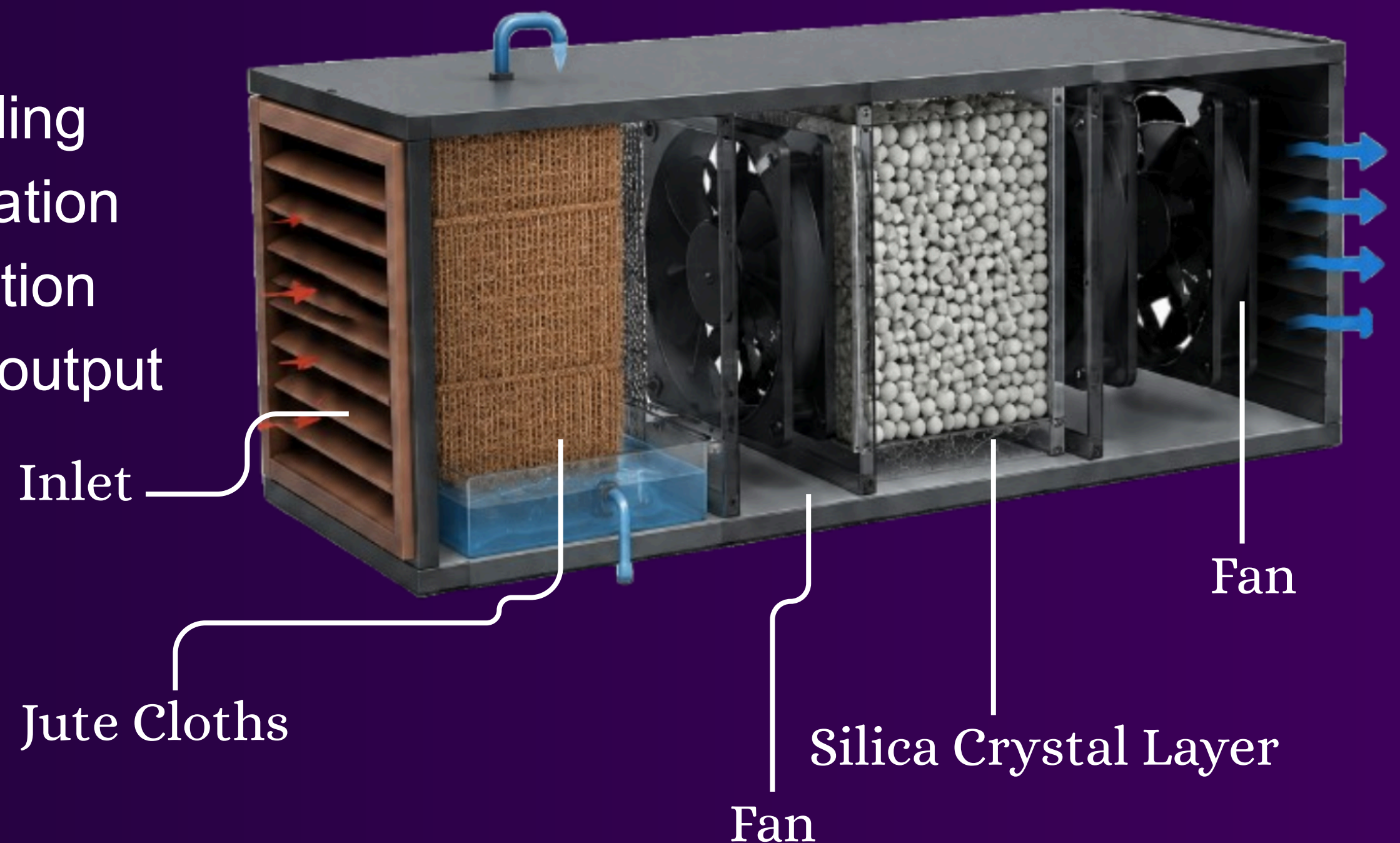
- A hybrid passive cooling prototype integrating evaporative cooling and natural materials.
- Multi-stage cooling using wetted gunny bag media and hydrogel-based moisture regulation.
- Real-time temperature monitoring using inlet and outlet sensors.





How VAYU Works

- ▶ **Step 1:** Warm air intake
- ▶ **Step 2:** Evaporative cooling
- ▶ **Step 3:** Forced air circulation
- ▶ **Step 4:** Humidity adsorption
- ▶ **Step 5:** Comfortable air output



Performance Metrics

Data derived from prototype testing phases.

3

°C Temp Reduction

80

% Less Power Use

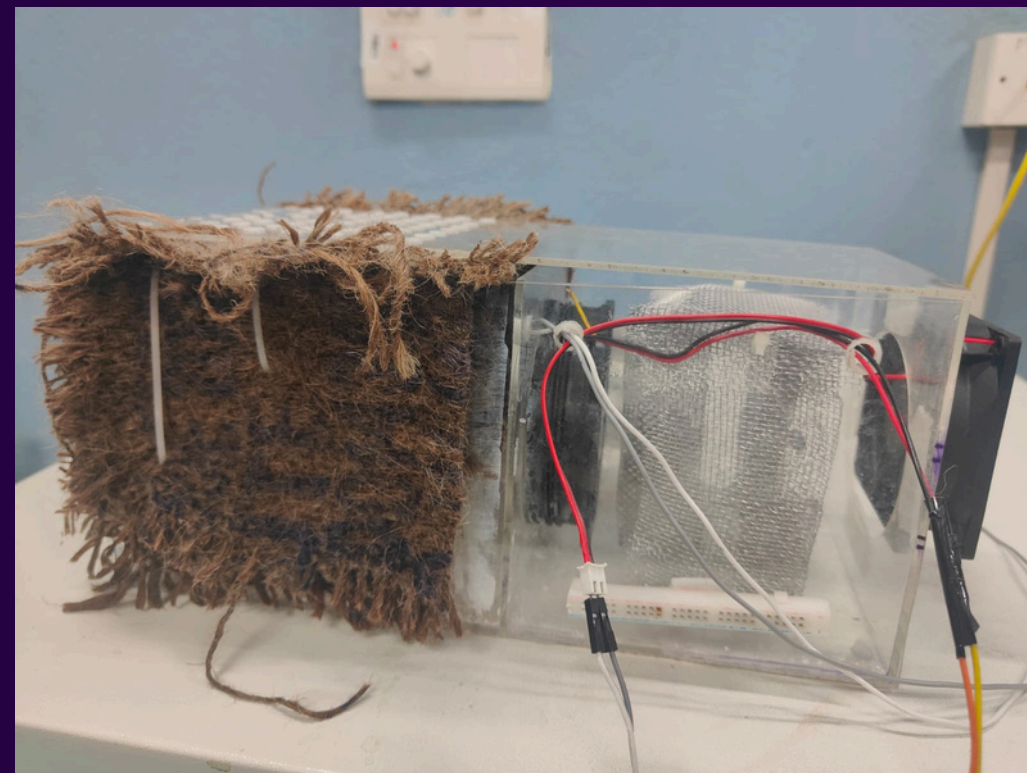
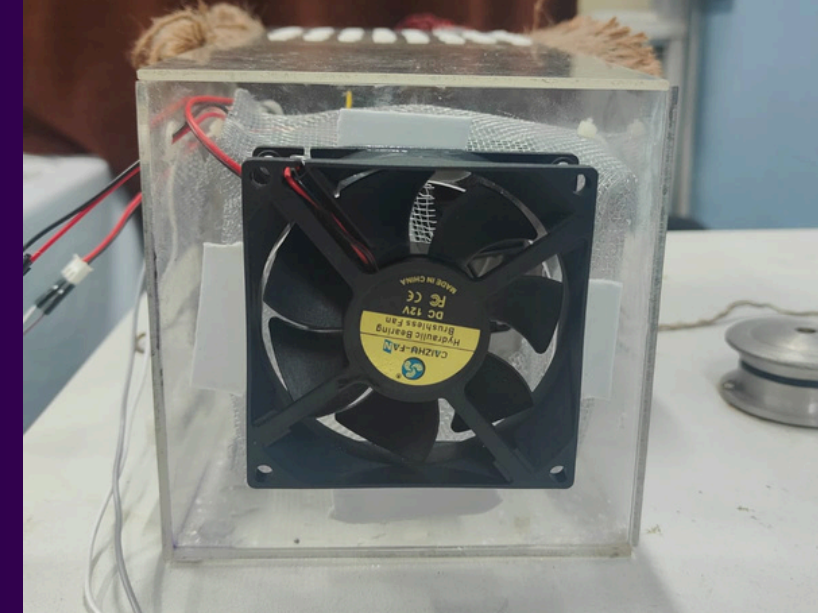
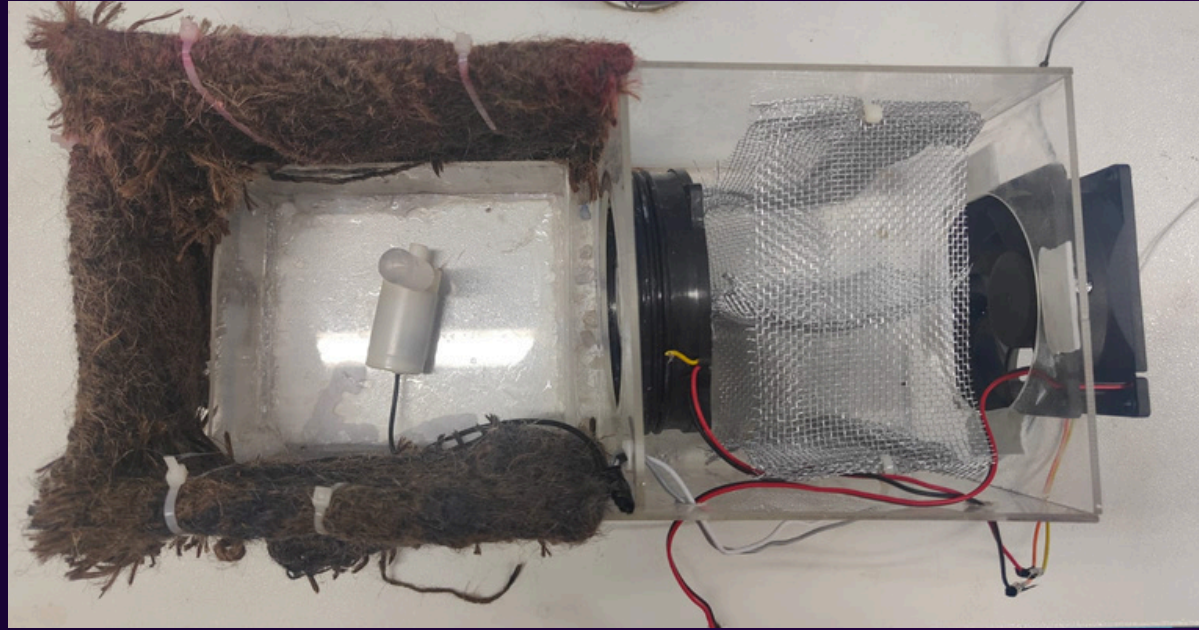
95

% Eco-Friendly

70

% Cost Savings

Real Images Of Prototype



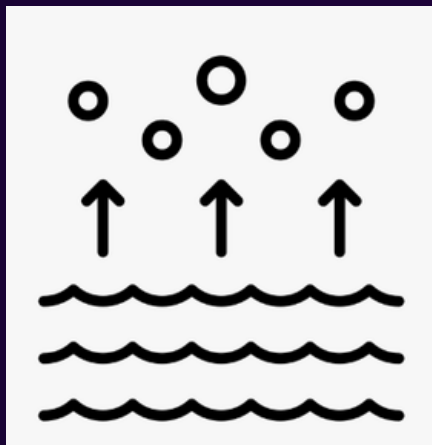
Scientific Principles Involved

Evaporative Cooling - Water absorbs heat while evaporating, lowering air temperature.

Adsorption - Silica gel captures moisture from air, reducing humidity.

Heat Transfer - Passive Cooling - Uses natural cooling processes with minimal energy input.

Airflow Dynamics - Fans ensure efficient circulation of cooled air.



VAYU
Crafted with



For Indians By Indians...

Thank You